

REQUIREMENTS DOCUMENT (LIGHT BRD)

Business objectives

Improve forecast accuracy to at least 90 percent at total level.
Identify regions and products that consistently miss targets.
Provide a 6-month revenue forecast with scenarios.

User stories

As a CFO, I want a 6-month revenue forecast with best and worst cases so that I can plan the budget.
As a Head of Sales, I want target achievement by region and product so that I can coach underperforming teams.
As a Demand Planner, I want to see seasonal patterns so that I can plan inventory for peak months.
As a Marketing Head, I want revenue and marketing spend side by side so that I can move spend to better-performing areas.

Functional requirements

Filter by year, quarter, region, product category and customer segment.
Show forecast versus actual revenue by month.
Show variance and target achievement by region and product.
Show the model forecast for the next 6 months.

KPI DICTIONARY

Total Revenue = sum of Actual Revenue.
Revenue Growth % = (Current period revenue minus previous period revenue) divided by previous period revenue.
Reported year over year.
Revenue Variance % = (Actual Revenue minus Forecasted Revenue) divided by Forecasted Revenue. Positive means actual beat the forecast.
Forecast Accuracy % = 1 minus (sum of absolute errors divided by sum of actual revenue). Also called 1 minus WAPE.
Forecast Bias % = (sum of forecast minus sum of actual) divided by sum of actual. Positive means the forecast is too optimistic.
Profit = Revenue minus Cost. Profit Margin % = Profit divided by Revenue.
Target Achievement % = Actual Revenue divided by Sales Target.
Average Revenue per Order = Total Revenue divided by number of orders. (The dataset has orders, not unique customers, so this stands in for revenue per customer. State this assumption openly.)
Marketing ROI = Revenue divided by Marketing Spend.

Assumptions

Actual Revenue equals Units Sold times Unit Price.
Cost includes product and delivery cost, excluding marketing.
Seasons: Winter (Jan-Feb), Summer (Mar-Jun), Monsoon (Jul-Sep), Festive (Oct-Dec).

Risks

Small sample size, so forecasts are indicative.
Only two years of history, so seasonality estimates are rough.